

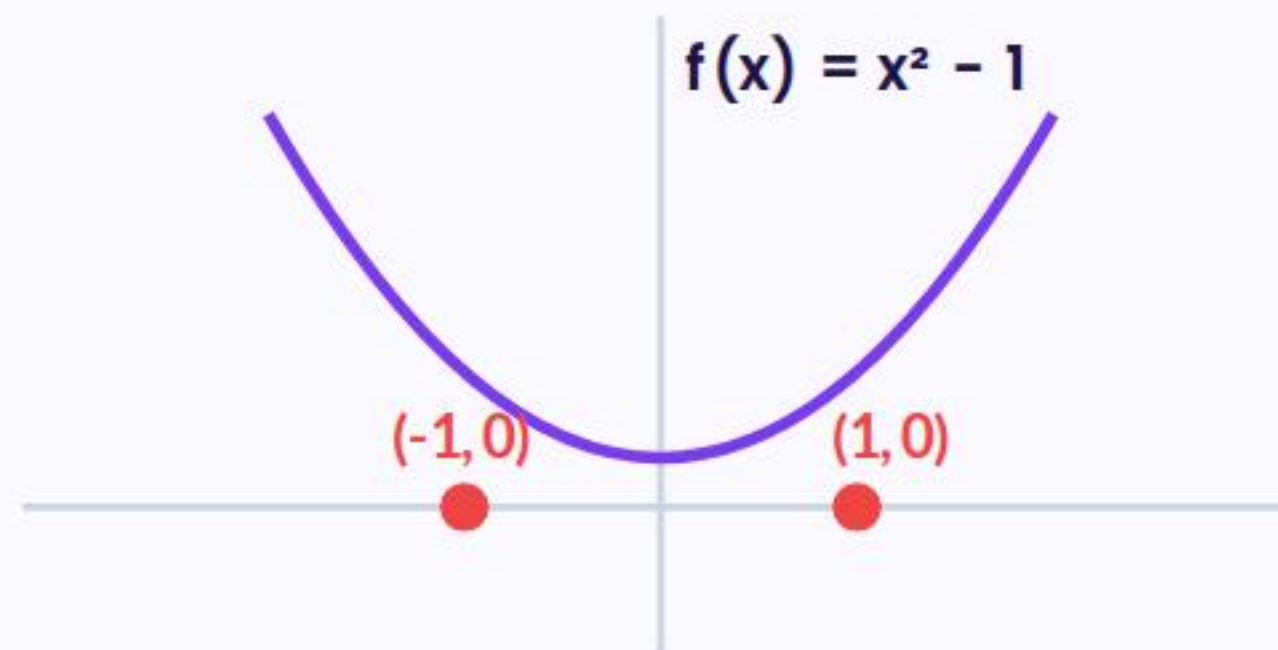
Chapter 2: **Complex Numbers**

Analytical foundations, geometric origins, and the algebraic rules of the imaginary system

REAL VS IMAGINARY DILEMMA

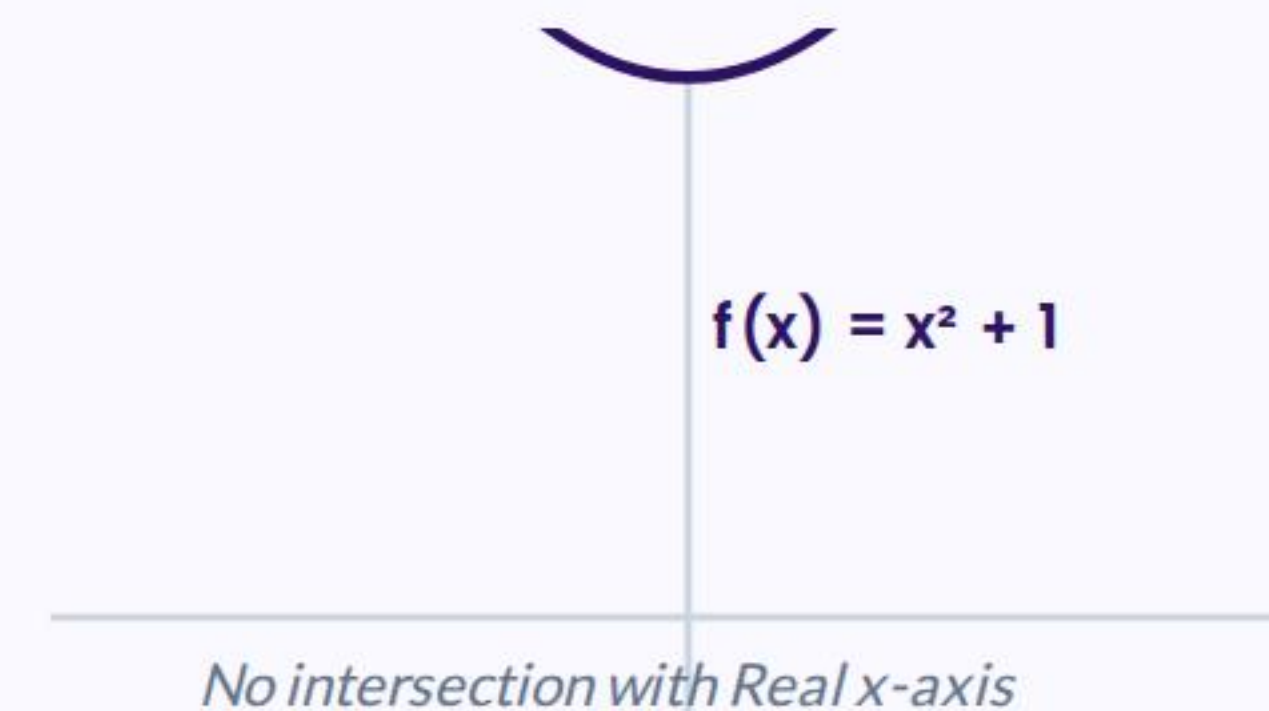
Equation 1: Real Solutions

Solving $X^2 - 1 = 0$ yields real roots $X = \pm 1$. Geometrically, the curve intersects the real axis.



Equation 2: No Real Solutions

Solving $X^2 + 1 = 0$ requires $X = \pm \sqrt{-1}$. Because the graph has no x-intercepts, we must create a new domain.



DEFINING IMAGINARY UNIT I

The Birth of Imaginary Numbers

When squaring any real number, the result is always non-negative ($y^2 \geq 0$). Thus, solving square roots of negative numbers requires a defined symbol.

Italian mathematician **Rafael Bombelli** recognized that we must introduce an imaginary unit, denoted by i :

$$i = \sqrt{-1}$$

The Core Axiom

- ✓ The primary defining property of the unit is:
 $i^2 = -1$
- ✓ It allows us to express the square root of any negative real number as:
 $\sqrt{-y^2} = i |y|$
- ✓ This bridges the gap between pure algebra and real analytical space.

PERIODIC POWERS OF I

Positive Index Cycles

Powers of i repeat sequentially every four terms:

Index	Simplification	Value
i^1	i	i
i^2	-1	-1
i^3	$i^2 \cdot i$	$-i$
i^4	$i^2 \cdot i^2$	1

Negative Indices & General Division

The division algorithm tells us that for any integer power i^n , we can write $n = 4q + k$ ($0 \leq k < 4$):

$$i^n = i^{4q+k} = (i^4)^q i^k = i^k$$

Reciprocal	Simplification	Value
i^{-1}	$\frac{1}{i} \cdot \frac{i}{i}$	$-i$
i^{-2}	$\frac{1}{-1}$	-1

KEY CONCEPTUAL CAVEATS

Consecutive Power Property

The sum of any four consecutive integral powers of the imaginary unit i is always zero:

$$i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0 \quad \forall n \in \mathbb{Z}$$

This follows directly because any consecutive set cycles through the elements $\{i, -1, -i, 1\}$, which sum to zero.

The Radical Caveat

The algebraic identity $\sqrt{ab} = \sqrt{a}\sqrt{b}$ is valid **only if at least one of the variables is non-negative**.

The Classic Contradiction: If both are negative:

$$\sqrt{-4}\sqrt{-9} = (2i)(3i) = 6i^2 = -6$$

However, calculating simultaneously:

$$\sqrt{(-4)(-9)} = \sqrt{36} = 6$$

Assuming unrestricted application leads to $6 = -6$, which is mathematically invalid.